Analytica Haven Data Analytics Curriculum (Feb–May 2026)

Duration: 12 Weeks (3 months)

Frequency: 3 classes per week, 2 hours each

Progression: Beginner → Intermediate → Professional/Expert

1. Week 1 – Beginner Level

Topics: Foundations of Data Analysis, Excel Basics, Descriptive Statistics

Objectives:

1.1. Understand the role and process of data analysis.

1.2. Navigate Excel confidently and apply basic formulas.

1.3. Compute descriptive statistics using Excel functions.

Expected Outcomes:

1.1. Define data analysis and its importance.

1.2. Perform basic calculations (SUM, AVERAGE, COUNT).

1.3. Apply statistical measures (mean, median, mode, variance, standard deviation).

Lesson Notes:

1.1. Tutors emphasize real-world examples (e.g., sales data).

1.2. Exercises: calculate average monthly sales, find variance in customer ratings.

1.3. Discussion: why descriptive statistics matter in business decision-making.

2. Week 2 – Beginner Level

Topics: Data Cleaning, PivotTables, Probability Basics

Objectives:

2.1. Learn Excel data cleaning techniques.

2.2. Build and interpret PivotTables.

2.3. Understand probability concepts and conditional probability.

Expected Outcomes:

2.1. Clean messy datasets.

2.2. Create PivotTables for summarizing data.

2.3. Apply probability rules to real-world scenarios.

Lesson Notes:

2.1. Tutors guide exercises on cleaning customer datasets.

2.2. PivotTable practice: group sales by region, calculate averages.

2.3. Probability example: likelihood of customer churn.

3. Week 3 – Intermediate Level

Topics: Advanced Excel Formulas, Inferential Statistics

Objectives:

3.1. Master lookup and logical functions.

3.2. Understand hypothesis testing and confidence intervals.

3.3. Perform t-tests using Excel's Data Analysis Toolpak.

Expected Outcomes:

3.1. Use VLOOKUP, INDEX-MATCH, and nested IFs.

3.2. Conduct hypothesis testing with real datasets.

3.3. Interpret confidence intervals in business contexts.

Lesson Notes:

3.1. Tutors emphasize practical lookup exercises (e.g., product catalog searches).

3.2. Case study: test if a marketing campaign increased sales.

3.3. Group discussion: interpreting p-values in decision-making.

4. Week 4 – Intermediate Level

Topics: SQL Basics – Core Queries & Data Extraction

Objectives:

4.1. Understand relational databases.

4.2. Write basic SQL queries (SELECT, WHERE, ORDER BY).

4.3. Apply filtering and limiting results.

Expected Outcomes:

4.1. Query databases for specific information.

4.2. Extract and filter data efficiently.

Lesson Notes:

4.1. Tutors demonstrate querying a customer database.

4.2. Exercises: retrieve top 10 customers by revenue.

4.3. Discussion: why SQL is essential for analysts.

5. Week 5 – Intermediate Level

Topics: SQL Joins, Aggregation & Grouping

Objectives:

5.1. Understand relationships between tables.

5.2. Perform joins and aggregations.

5.3. Use GROUP BY and HAVING clauses.

Expected Outcomes:

5.1. Combine multiple tables.

5.2. Summarize data using aggregate functions.

Lesson Notes:

5.1. Tutors guide join exercises (e.g., customers + orders).

5.2. Case study: calculate average order value per region.

5.3. Team exercise: group data by product category.

6. Week 6 – Intermediate Level

Topics: SQL Subqueries, Window Functions, Integration

Objectives:

6.1. Write subqueries in different contexts.

6.2. Apply window functions for ranking and comparisons.

6.3. Export SQL results to Excel.

Expected Outcomes:

6.1. Use ROW_NUMBER, RANK, LAG/LEAD.

6.2. Perform advanced SQL analysis.

Lesson Notes:

6.1. Tutors emphasize ranking exercises (e.g., top 5 products).

6.2. Case study: compare customer purchases month-over-month.

6.3. Integration: export SQL results for Excel analysis.

7. Week 7 – Professional Level

Topics: Power BI Introduction & Data Modeling

Objectives:

7.1. Understand Power BI ecosystem.

7.2. Connect to multiple data sources.

7.3. Build relationships between tables.

Expected Outcomes:

7.1. Load and transform data in Power BI.

7.2. Create star schema models.

Lesson Notes:

7.1. Tutors demonstrate connecting SQL + Excel datasets.

7.2. Exercises: build a simple star schema.

7.3. Discussion: why data modeling matters for dashboards.

8. Week 8 – Professional Level

Topics: DAX Basics & Calculated Columns

Objectives:

8.1. Learn DAX fundamentals.

8.2. Differentiate between calculated columns and measures.

8.3. Apply common DAX functions.

Expected Outcomes:

8.1. Write DAX formulas.

8.2. Use time intelligence functions for reporting.

Lesson Notes:

8.1. Tutors emphasize CALCULATE and FILTER examples.

8.2. Case study: calculate year-to-date sales.

8.3. Group exercise: build measures for KPIs.

9. Week 9 – Professional Level

Topics: Power BI Visualizations & Dashboard Design

Objectives:

9.1. Choose appropriate visuals for data.

9.2. Customize and format dashboards.

9.3. Add interactivity with slicers and filters.

Expected Outcomes:

9.1. Design professional dashboards.

9.2. Publish dashboards to Power BI Service.

Lesson Notes:

9.1. Tutors guide exercises on dashboard design.

9.2. Case study: build a customer churn dashboard.

9.3. Team activity: critique and improve dashboard designs.

10. Week 10 – Professional Level

Topics: Integrated Project – End-to-End Analysis

Objectives:

10.1. Apply SQL, Excel, and Power BI in one workflow.

10.2. Solve a real-world business problem.

10.3. Present findings effectively.

Expected Outcomes:

10.1. Complete an end-to-end analysis project.

10.2. Deliver a 5-minute presentation.

Lesson Notes:

10.1. Tutors mentor project teams.

10.2. Example scenario: investigate customer churn.

10.3. Emphasis on storytelling with data.

11. Week 11 – Expert Level

Topics: Capstone Project – Advanced Business Case

Objectives:

11.1. Work on a new dataset and business case.

11.2. Apply advanced SQL, Excel, and Power BI skills.

11.3. Prepare professional deliverables.

Expected Outcomes:

11.1. Produce SQL scripts, Excel analysis, and Power BI dashboards.

11.2. Deliver a polished presentation.

Lesson Notes:

11.1. Tutors act as project advisors.

11.2. Encourage collaboration and peer review.

11.3. Focus on professional presentation skills.

12. **Week 12 – Expert Lev**el

Topics: Career Preparation & Final Capstone Delivery

Objectives:

12.1. Finalize capstone projects.

12.2. Practice career-oriented presentations.

12.3. Prepare for job interviews and portfolio building.

Expected Outcomes:

12.1. Present final projects.

12.2. Build a portfolio showcasing SQL, Excel, and Power BI expertise.

Lesson Notes:

12.1. Tutors emphasize communication and storytelling.

12.2. Mock interviews and feedback sessions.

12.3. Collaborative reflection on learning journey.